

# assignment\_kine1D\_graphing

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:52pm

## Quiz Instructions

2 attempts. There is no tutorial.



### Question 1 10 pts

An object is moving with constant non-zero velocity in the  $+x$  direction. The velocity versus time graph of this object is

☐

a vertical straight line.

☐

a parabolic curve.

☐

a horizontal straight line.

☐

a straight line making an angle with the time axis.



### Question 2 6.7 pts

An object is moving with constant non-zero acceleration in the  $+x$  direction. The position versus time graph of this object is

☐

a horizontal straight line.

☐

a straight line making an angle with the time axis.

☐

a parabolic curve.

☐

a vertical straight line.



### Question 3 6.7 pts

The slope of a velocity versus time graph gives

☐

displacement.

☐

velocity.



acceleration.



the distance traveled.



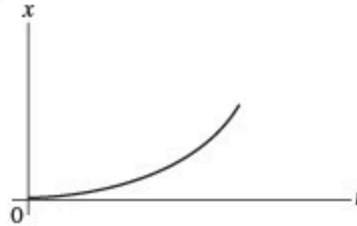
Question 4 6.7 pts

Which of the following graphs represent an object at rest? (There could be more than one correct choice.)

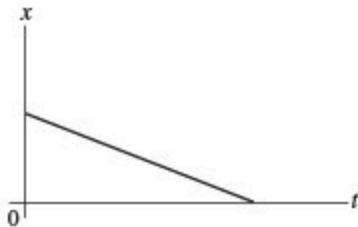
(a)



(b)



(c)



(d)



(e)



graph d



graph b



graph a



graph c

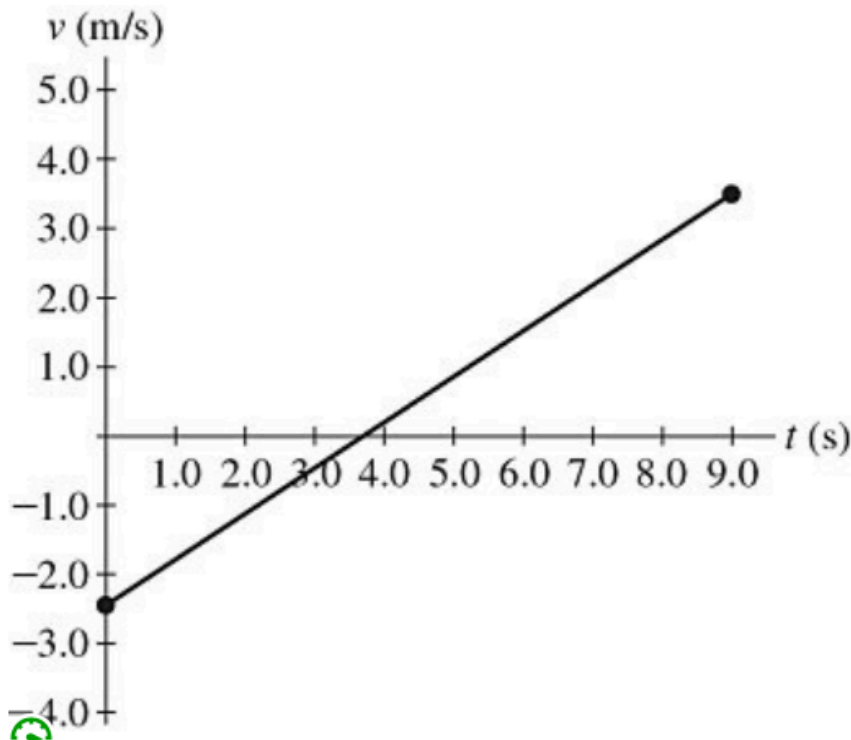


graph e



## Question 5 10 pts

The motion of a particle is described in the velocity vs. time graph shown in the figure.



Over the nine-second interval shown, we can say that the *speed* of the particle

hint: before 3.5s , the velocity is negative so its moving to the left. But the magnitude of the velocity is decreasing. After 3.5 s, the velocity is positive. so its moving @ right. But the magnitude is increasing.

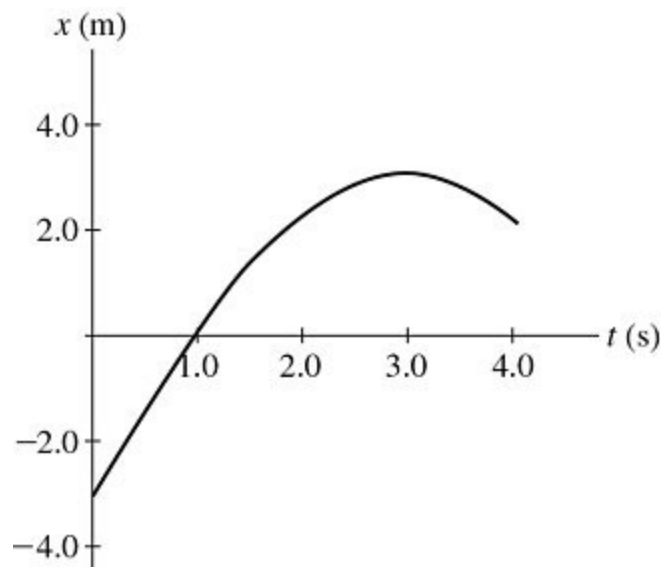
So it reversed its direction.

- ☐ decreases and then increases.
- ☐ only decreases.
- ☐ remains constant.
- ☐ only increases.
- ☐ increases and then decreases.



## Question 6 10 pts

The graph in the figure shows the position of a particle as it travels along the x-axis.



At what value of  $t$  is the speed of the particle equal to 0 m/s?

hint: the slope is 0 (horizontal tangent)

☐

2 s

☐

1 s

☐

4 s

☐

0 s

☐

3 s

⋮

Question 7 6.7 pts

If the velocity versus time graph of an object is a horizontal line, the object is

☐

moving with increasing speed.

☐

at rest.

☐

moving with constant non-zero acceleration.

☐

moving with zero acceleration.

⋮

Question 8 6.7 pts

The slope of a position versus time graph gives

☐

displacement.

☐

velocity.

☐

the distance traveled.

☐

acceleration.



#### Question 9 6.7 pts

If the position versus time graph of an object is a horizontal line, the object is

☐

moving with constant non-zero speed.

☐

moving with constant non-zero acceleration.

☐

at rest.

☐

moving with increasing speed.



#### Question 10 10 pts

An object is moving with constant non-zero velocity in the  $+x$  direction. The position versus time graph of this object is

☐

a straight line making an angle with the time axis.

☐

a parabolic curve.

☐

a vertical straight line.

☐

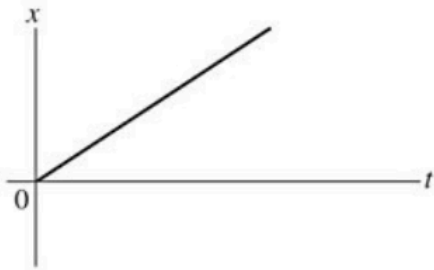
a horizontal straight line.



#### Question 11 6.7 pts

Which of the following graphs represent an object having zero acceleration?

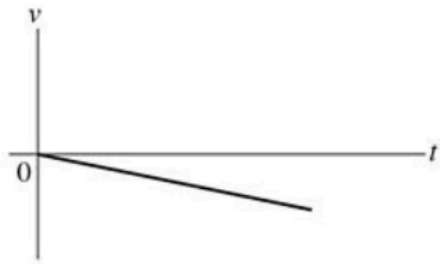
(a)



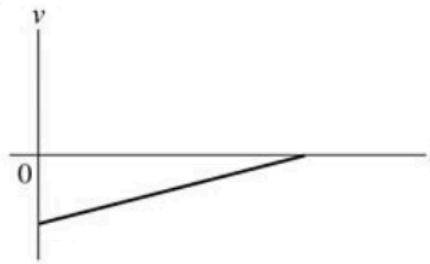
(b)



(c)



(d)



☐ graphs a and b



only graph b



only graph a



graphs b and c



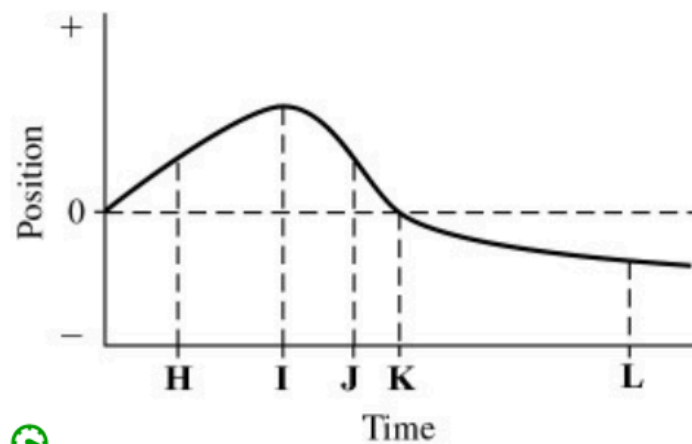
graphs c and d



Question 12 6.7 pts

Use Figure 2.1 to answer the following question(s).

Figure 2.1



The graph in Figure 2.1 shows the position of an object as a function of time. The letters H-L represent particular moments of time. At which moment in time is the speed of the object the highest?

hint: the highest speed is at the point where the slope is the largest. "the sloppier" tangent. (don't worry about sign)

☐

L

☐

I

☐

K

☐

J

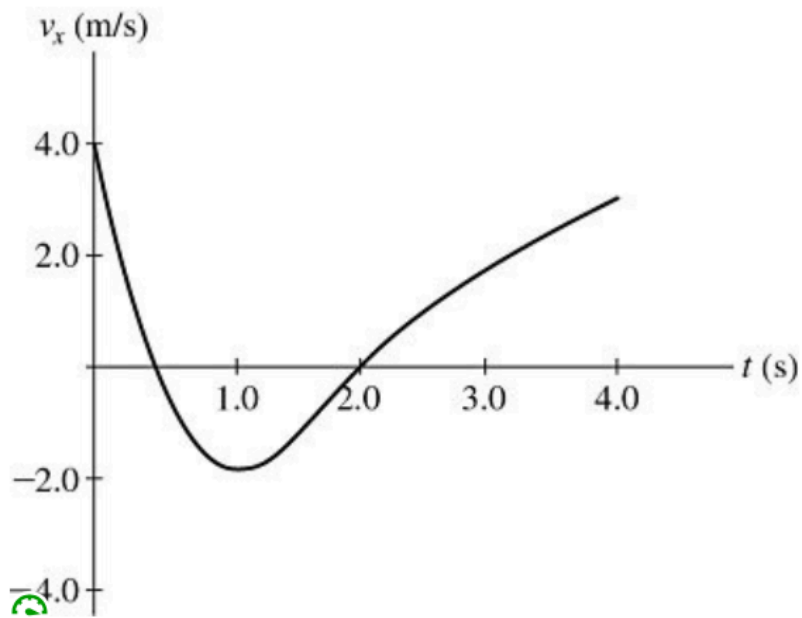
☐

H

☐

Question 13 6.7 pts

The graph in the figure represents the velocity of a particle as it travels along the  $x$ -axis. What is the average acceleration of the particle between  $t = 2.0$  s and  $t = 4.0$  s?



hint: acceleration is the change in velocity over the change in time.

☐

-2.0 m/s<sup>2</sup>

☐

0

☐

1.5 m/s<sup>2</sup>

☐

3.0 m/s<sup>2</sup>

Not saved

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